

Program 6

Design and implement C/C++ Program to solve 0/1 Knapsack problem using Dynamic Programming method.

```
#include<stdio.h>

int v[20][20];

int max1(int a,int b)
{
    return(a>b)?a:b;
}

void main()
{
    int i, j, w[20], p[20], n, max;
    printf("enter the number of items\n");
    scanf("%d", &n);
    for(i=1; i<=n; i++)
    {
        printf("enter the weight & profit of the item %d:",i);
        scanf("%d%d", &w[i],&p[i]);
    }

    printf("enter the capacity of knapsack");
    scanf("%d", &max);
    for(i=0; i<=n; i++)
        v[i][0]=0;
    for(j=0;j<=max;j++)
        v[0][j]=0;
    for(i=1;i<=n;i++)
        for(j=1;j<=max; j++)
        {
            if(w[i]>j)
                v[i][j]=v[i-1][j];
            else
                v[i][j]=max1(v[i-1][j],v[i-1][j-w[i]]+p[i]);
        }

    printf("\n\n the table is\n");
    for(i=0; i<=n; i++)
    {
        for(j=0; j<=max; j++)
            printf("%d\t",v[i][j]);
        printf("\n");
    }

    printf("\n the maximum profit is %d",v[n][max]);
    printf("\n the most valuable subset is: {");
    j=max;
    for(i=n;i>=1;i--)
        if(v[i][j]!=v[i-1][j])
        {
            printf("\t item %d",i);
            j=j-w[i];
        }
}
```

```
    }  
    printf("{}");  
}
```

Sample Input and Output:

enter the number of items

4

enter the weight & profit of the item 1:2 12

enter the weight & profit of the item 2:1 10

enter the weight & profit of the item 3:3 20

enter the weight & profit of the item 4:2 15

enter the capacity of knapsack5

the table is

0	0	0	0	0	0
0	0	12	12	12	12
0	10	12	22	22	22
0	10	12	22	30	32
0	10	15	25	30	37

the maximum profit is 37

the most valuable subset is:{ item 4 item 2 item 1}